

KAGGLE

Antimicrobial Resistance Prediction from MALDI-TOF

Authors

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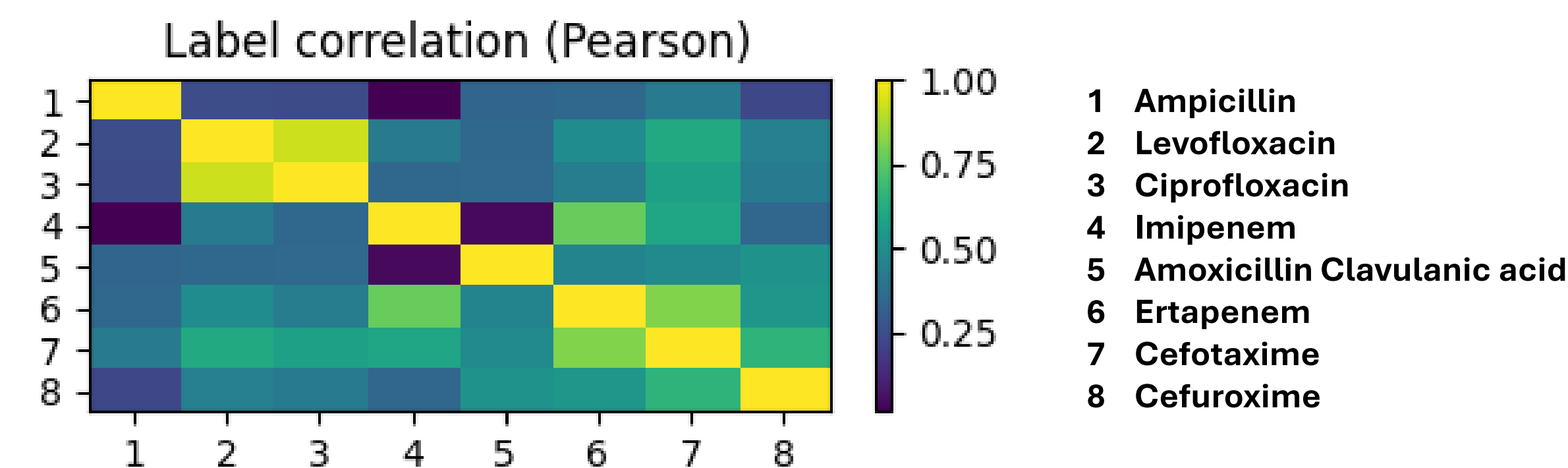
INTRODUCTION

Multi-label outputs: correlated resistance predictions across antibiotics.

Incomplete supervision: training data contains missing labels.

Species heterogeneity: resistance mechanisms differ across bacterial species.

Non-linear structure: complex interactions between spectra, species, and resistance.



Semi-supervised (PSEUDO-LABELING)

MOTIVATION

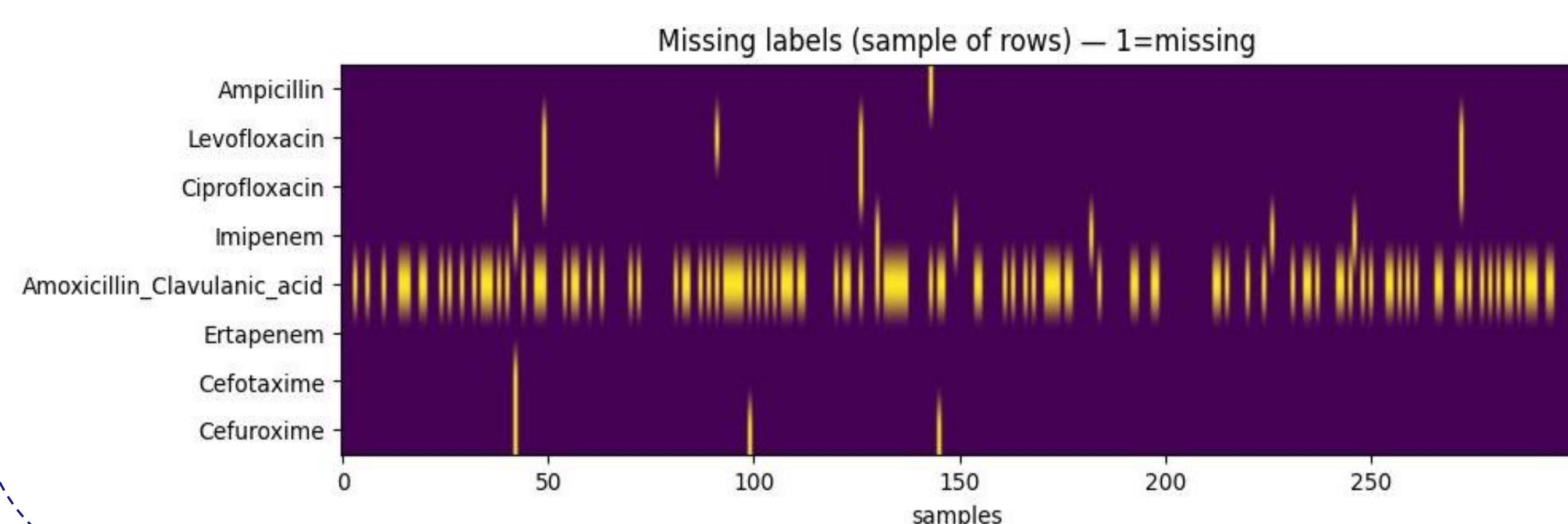
- ~43% of labels missing for Amox/Clav

MODEL

- Add high-confidence synthetic labels for Amox/Clav

RESULTS

- Downgrades AUC for all targets



SIMPLE MACHINE LEARNING AND DEEP LEARNING MODELS

RANDOM FORESTS

- Random Forest/antibiotic
- Used as baseline

CNN

- LocallyConnected1D
- Species embeddings
- Multi-head (1/antibiotic)

LIGHTGBM

- Per antibiotic
- Higher variance
- Diversity source.

STACKING AND META LEARNING

MOTIVATION

- Resistance patterns correlated
- Meta-learning for information transfer + generalization
- Weak AUC seen in Amox/Clav

MODEL

- Species-aware XGBoost
- For Amox/Clav used Ampicillin, Cefotaxime and Cefuroxime OOF predictions
- XGBoost + LR

RESULTS

- Higher AUC for Amox/Clav
- Less variance
- Better generalization
- Better calibration

FEATURE ENGINEERING AND SVM

MOTIVATION

- Local spectral shifts
- Geometry aware modeling

MODEL

- Nyström kernel approximation
- Linear SVM

RESULTS

- Gains only for Ampicillin
- Used as specialist in ensemble

XGBOOST

MOTIVATION

- MALDI spectra are high-dimensional and continuous
- Useful interactions are not known a priori

MODEL

- 1 XGBoost per antibiotic
- Models per species for Amox/Clav

RESULTS

- High AUC for carbapenems & cephalosporins

CATBOOST

MOTIVATION

- Non-linear feature interactions
- Categorical species is informative

MODEL

- 1 CatBoost per antibiotic
- Models per species for Amox/Clav

RESULTS

- Best global ranking performance
- High AUC for carbapenems & cephalosporins

ENSEMBLES AND BLENDING

MOTIVATION

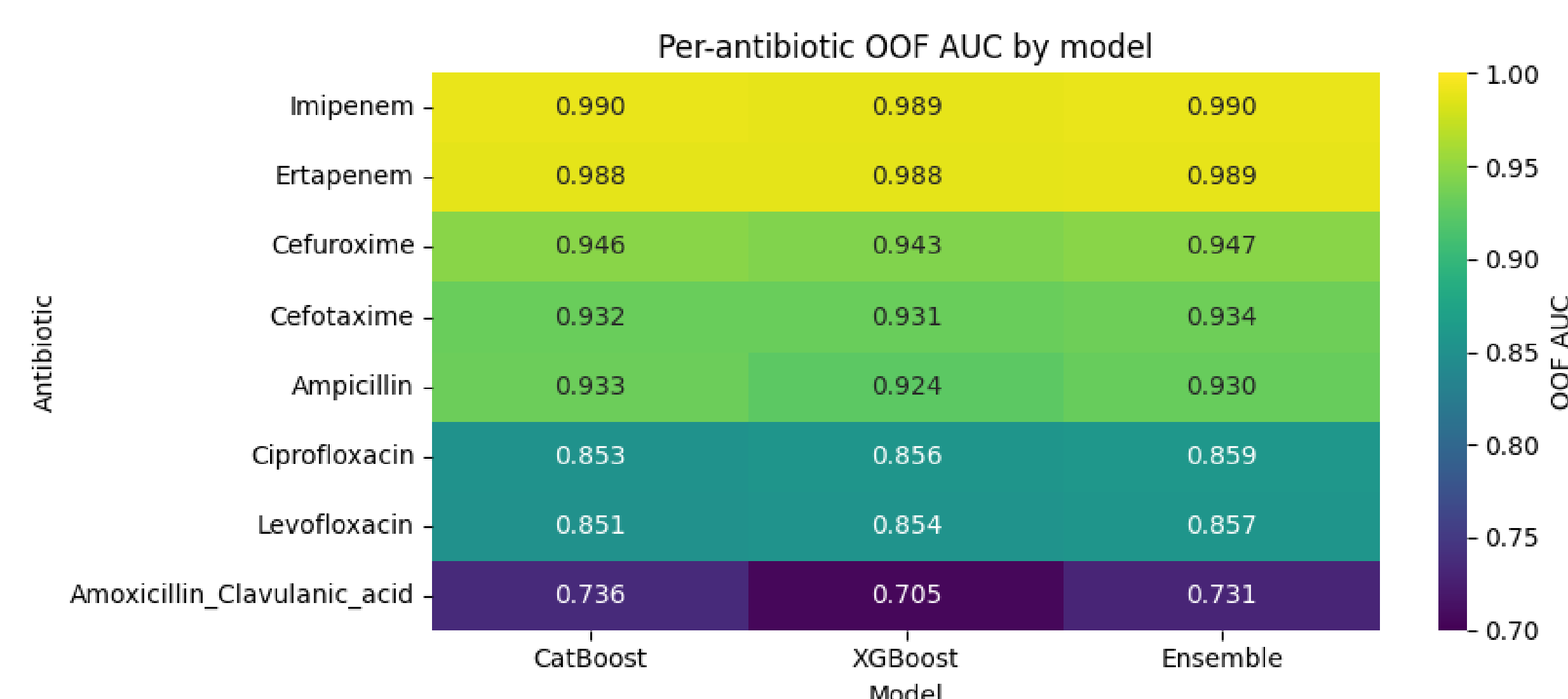
- No best model for all antibiotics
- Models differ in inductive bias
- Combining improves robustness under imbalance

MODEL

- XGB, CatBoost, LGBM, SVM specialists
- Rank averaging to remove calibration effects

RESULTS

- Consistent AUC improvement
- Reduced variance across antibiotics
- Robust final predictions



CONCLUSION AND REMARKS

- Catboost outperformed** individual XGBoost models
- Simple ensembles** > complex stacking
- Model performance dependence on the antibiotic: **carbapenems easy to predict** (near-ceiling AUCs), **heterogeneous β -lactams required specialist models and ensembling**